

Terminal Gaussian jet threshold, floor-stable fixed-law action, and conditional gamma-7/12 aggregate-collar boundary

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claim A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION

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1. Purpose and scope

R-133 isolated three possible successors: a blockwise score theorem, a direct fixed-law estimate, and an aggregate positive-exponent collar estimate. R-134 tests all three against the complete six-row current and the existing root/visit/forest ownership.

The outcome is a real advance but not production closure.

1. The last independent six-real Gaussian source block in a fixed R-104 chart can regularise the rational coefficient without an inverse feedback Jacobian, provided its pointwise covariance has an added uniform full-rank lower bound. Sharp six-dimensional negative moments control the second and third quotient jets. The fourth jet is critical and has an explicit logarithmic L^2 divergence.
2. The density floor can be removed at the action level before any differentiation. The complete Pauli sum improves the naive rowwise constant:

$$\|C_{6,e}(w)v - C_{6,0}(w)v\| \leq \alpha\sqrt{c_1e}|v|, \quad \alpha^2c_1 = \frac{3}{320P}. \quad (1.1)$$

After all owners are recombined, one Hilbert direct-sum inequality has no root or visit multiplicity. It reduces the action problem to one signed forest-minus-weighted-current form.

3. The weighted current energy cannot then be paid separately by source and sextic. The uncontrolled covariance-normal Gaussian has bounded value moments but linearly divergent gradient energy. Separating the forest therefore destroys the Wick cancellation which the forest was built to keep.
4. A conditional root-atom estimate with spatial exponent $s > 7/12$ does sum to the accepted $\mathcal{B}_{7/12}^{\text{out}}$ ledger with an exact geometric constant. At $s = 2/3$ the square constant is $0.658881625872614\dots$. A direct fixed-collar form has the smaller constant $0.285928885855479\dots$. Neither conditional estimate supplies its own production atom or one-use q_k ledger.

The result is T4 at fixed finite cutoff and positive floor. It does not prove a terminal covariance lower bound uniform across cutoffs, charts, subdivisions, or refinements; an owner-preserving terminal disintegration; the complete signed forest-current inequality; the production one-use atom hypotheses; a collar-uniform near/balanced headroom; the low matrix, matching energy, strict gap, or absolute anchor; either A13 gate; OVERLAP_{src}; Nelson; removals; an interacting measure; or Sector-A closure.

2. Complete-owner firewall

Fix one finite temporally faithful triangular chart, one positive density floor, and the common legal heat. Recombine all representation-preserving same-root visits before estimating. Sum an exhaustive contraction-closed orthogonal output family and retain the trace covariance, conditional future variance, adapted R-063 forest, rational rows, regular Cartan block, live 2680/729 rational curl, low owner, source, and terminal sextic exactly once.

R-125 gives its compact bridge only after the exhaustive output-cluster sum. Thus let o denote a recombined root-spatial owner, let $\mathcal{C}$ range over its contraction-closed orthogonal output family, and put

$$\Psi_o = \Delta \mathcal{V}_o - \Delta F_{063,o}^{\text{ad}}. \quad (2.1)$$

Equivalently, before taking a Hessian,

$$\boxed{\begin{aligned} \mathcal{P}_{\text{comp},o} &:= \sum_{\mathcal{C}} \mathcal{P}_{\text{comp},o,\mathcal{C}} = \frac{1}{2} F_{063,o}^{\text{ad}} - \frac{1}{2} \text{Var}_f(X_o), \\ X_o &:= (P_{\mathcal{C}} C_{6,e_o,\mathcal{C}}(W_{o,\mathcal{C}}) V_{o,\mathcal{C}})_{\mathcal{C}} \in \bigoplus_{\mathcal{C}} \mathcal{H}_{\mathcal{C}}. \end{aligned}} \quad (2.2)$$

No forest is allocated to a single cluster. Orthogonality makes the direct- sum variance in (2.2) exactly the exhaustive cluster sum. The variance and forest are alternate coordinates for the same primitive trace owner, not two independent reserves. Same-root visit subdivision preserves (2.2) and creates no new covariance or forest owner.

The action source and terminal sextic are appended only after summing (2.2). The source stays outside the physical predictable pullback. No statement in this note moves it inside or spends it twice.

3. Sharp six-row floor decomposition

Write $w = (u, \chi) \in \mathbb{R}^6$, $r = |w|^2$, and let S_A be the three Pauli generators acting on the complex doublet. Put

$$m_A(w) = w^T S_A w, \quad q_{A,e}(w) = \frac{m_A(w)}{r+e}, \quad \alpha = \frac{5}{9}. \quad (3.1)$$

The three raw rows are independent of e . The three rational rows are

$$L_{A,e}(w; v) = 2 \langle (S_A - \alpha q_{A,e} I) w, v \rangle. \quad (3.2)$$

Define $C_{6,0}$ by replacing $(r+e)^{-1}$ by r^{-1} for $r > 0$ and by the continuous value at $w = 0$. Then $R_e = C_{6,e} - C_{6,0}$ has only rational rows, and

$$(R_e(w)v)_A = 2\alpha\sqrt{c_1} \frac{e m_A(w)}{r(r+e)} \langle w, v \rangle. \quad (3.3)$$

If $s = |P_{\text{dbl}} w|^2$, Pauli completeness is

$$\sum_{A=1}^3 m_A(w)^2 = s^2. \quad (3.4)$$

Since $s \leq r$ and $|\langle w, v \rangle|^2 \leq r|v|^2$,

$$\begin{aligned} \|R_e(w)v\|^2 &= 4\alpha^2 c_1 e^2 \frac{s^2 |\langle w, v \rangle|^2}{r^2 (r+e)^2} \\ &\leq 4\alpha^2 c_1 e^2 \frac{r}{(r+e)^2} |v|^2 \leq \boxed{\alpha^2 c_1 e |v|^2}. \end{aligned} \quad (3.5)$$

The last inequality is $(r - e)^2 \geq 0$. Equality is attained on the active doublet with v parallel to w and $r = e$, so the constant is sharp within this algebra. Using the pinned coefficients,

$$\alpha^2 c_1 = \frac{3}{320P}, \quad \beta_{\text{op}} = 4(c_0 + c_1) = \frac{339}{2000P}, \quad (3.6)$$

and the accepted zero-floor envelope is

$$\|C_{6,0}(w)v\|^2 \leq \beta_{\text{op}}|w|^2|v|^2. \quad (3.7)$$

4. Root- and visit-free direct action inequality

Let o range over recombined root-spatial owners and $\mathcal{C}$ over each owner's exhaustive output family. Put

$$A^2 = \sum_{o,\mathcal{C}} \mathbb{E}|W_{o,\mathcal{C}}|^2 |V_{o,\mathcal{C}}|^2, \quad B_e^2 = \sum_{o,\mathcal{C}} e_{o,\mathcal{C}} \mathbb{E}|V_{o,\mathcal{C}}|^2, \quad \mathfrak{F} = \sum_o \mathbb{E} F_{063,o}^{\text{ad}}. \quad (4.1)$$

Apply Minkowski once in $L^2(\Omega; \oplus_{o,\mathcal{C}} \mathcal{H}_{\mathcal{C}})$, rather than cluster by cluster. Equations (3.5)–(3.7) give

$$\left(\sum_{o,\mathcal{C}} \mathbb{E}|C_{6,e_{o,\mathcal{C}}}(W_{o,\mathcal{C}})V_{o,\mathcal{C}}|^2 \right)^{1/2} \leq \sqrt{\beta_{\text{op}}}A + \alpha\sqrt{c_1}B_e. \quad (4.2)$$

Using $\text{Var } X_o \leq \mathbb{E}|X_o|^2$ in the exact post-cluster bridge (2.2) yields

$$\boxed{\sum_o \mathbb{E} \mathcal{P}_{\text{comp},o} \geq \frac{1}{2}\mathfrak{F} - \frac{1}{2} \left(\sqrt{\beta_{\text{op}}}A + \alpha\sqrt{c_1}B_e \right)^2}. \quad (4.3)$$

There is no owner-count, visit-count, or root-count factor in (4.3). For every $\vartheta > 0$ it also implies

$$\sum_o \mathbb{E} \mathcal{P}_{\text{comp},o} \geq \frac{1}{2}\mathfrak{F} - \frac{1+\vartheta}{2}\beta_{\text{op}}A^2 - \frac{1+\vartheta^{-1}}{2}\alpha^2 c_1 B_e^2. \quad (4.4)$$

This is a value/action theorem. It cannot be differentiated to produce the R-131 physical Hessian. Section 5 gives an exact bounded-action versus singular-Hessian warning.

5. Uniform finite-moment caps do not repair fixed-law curvature

Retain the R-132 scalar matched-trace ray

$$f_\delta(s) = s - \frac{5}{9} \frac{s^3}{s^2 + \delta^2}, \quad S \sim \frac{1}{2}(\delta_\delta + \delta_1). \quad (5.1)$$

Under common translation of the two atoms,

$$\mathcal{P}_{\text{comp}}''(0) = -4c_1 \{ \text{Var}(f'_\delta) + \text{Cov}(f_\delta, f''_\delta) \} = -\frac{10c_1}{81\delta} + O(1). \quad (5.2)$$

Nevertheless,

$$\mathbb{E}S = \frac{1+\delta}{2}, \quad \text{Var } S = \frac{(1-\delta)^2}{4}, \quad \mathbb{E}(S - \mathbb{E}S)^3 = 0, \quad \mathbb{E}|S|^p \leq 1 \quad (p > 0). \quad (5.3)$$

At the production floor $\delta = 10^{-6}$ the executable evaluation is

$$\mathcal{P}_{\text{comp}}''(0) = -937.498476563437\dots, \quad q_{\text{src}} + q_6 = \frac{9}{10} + \frac{9}{2}\mathbb{E}S^4 = \frac{63}{20} + \frac{9}{4}10^{-24}. \quad (5.4)$$

Thus 3.15 is only its displayed decimal to ordinary precision, not the exact value. Thus uniform upper caps on the mean magnitude, covariance, third moment, or any bounded finite list of positive moments, together with the displayed bounded source and sextic total, cannot give a floor-uniform fixed-law Hessian theorem. The exact mean in this family satisfies $\delta = 2\mathbb{E}S - 1$ and therefore identifies the floor parameter; no impossibility is claimed from the exact moment values as full data. This sharpens the scope of the registered R-132 law-free no-go; it is not a new production-law counterexample.

There is a precise one-dimensional density repair. Direct differentiation gives

$$\frac{3}{8} \leq f'_\delta \leq 1, \quad \|f''_\delta\|_{L^1(\mathbb{R})} = \frac{25}{18}, \quad \|sf''_\delta\|_{L^1(\mathbb{R})} = \frac{5\sqrt{3}}{6}\delta. \quad (5.5)$$

If S has density bounded by M , then

$$\text{Var}(f'_\delta) \leq \frac{25}{256}, \quad (5.6)$$

and

$$|\text{Cov}(f_\delta, f''_\delta)| \leq M \left(\frac{5\sqrt{3}}{6}\delta + \frac{25}{18}\mathbb{E}|S| \right). \quad (5.7)$$

Equations (5.6)–(5.7) identify anti-concentration, rather than another positive moment cap, as a viable curvature datum. The mean-dependent term still requires the source/sextic ledger.

6. Sharp six-real Gaussian negative moments

Let G be standard Gaussian in $\mathbb{R}^6$ and

$$X = m + \Gamma^{1/2}G, \quad \Gamma \succeq \lambda I_6, \quad \lambda > 0. \quad (6.1)$$

Theorem 6.1 (terminal Gaussian anti-concentration). For every $e \geq 0$ and $0 < q < 6$,

$$\mathbb{E}(e + |X|^2)^{-q/2} \leq C_{6,q}\lambda^{-q/2}, \quad C_{6,q} = \frac{\Gamma(3 - q/2)}{2^{1+q/2}}. \quad (6.2)$$

The constant is sharp as $e \downarrow 0$, $m = 0$, and $\Gamma = \lambda I_6$. In particular,

$$C_{6,2} = \frac{1}{4}, \quad C_{6,4} = \frac{1}{8}. \quad (6.3)$$

Proof. Write $\Gamma = \lambda I_6 + \Sigma$ and condition on the independent Gaussian with covariance Σ . The function $(e + |x|^2)^{-q/2}$ is radial and decreasing. Anderson's inequality, equivalently its layer-cake form for centred Gaussian balls, makes every conditional translation no larger than the centred $N(0, \lambda I_6)$ value. For $R = |G|$,

$$\mathbb{E}R^{-q} = 2^{-q/2} \frac{\Gamma((6 - q)/2)}{\Gamma(3)} = \frac{\Gamma(3 - q/2)}{2^{1+q/2}}. \quad (6.4)$$

This proves (6.2). Equality in the limiting centred isotropic case proves sharpness. $\square$

At $q = 6$ the Gamma factor has a pole. Radially, the squared critical jet has the integral $\int_0^1 r^5 r^{-6} dr$, hence logarithmic divergence.

7. Normalized quotient jet threshold

For $S = S^T$, $\|S\| \leq 1$, let

$$d_x = e + |x|^2, \quad n_x = \frac{x}{\sqrt{d_x}}, \quad F_S(x) = (n_x^T S n_x)x, \quad J_S = DF_S. \quad (7.1)$$

R-092 proves the polynomial identity $J_S = \mathcal{J}_S(n_x)$ and its secant bound. The derivative constants used here follow directly rather than being imported as unexplained numerals. For $q = n^T S n$, $P = n n^T$, and $\mathcal{J}_S = qI + 2PS - 2qP$, the bounds

$$\|Dq\| \leq 2, \quad \|D^2 q\| \leq 2, \quad \|DP\| \leq 2, \quad \|D^2 P\| \leq 2$$

give, term by term,

$$\|D\mathcal{J}_S\| \leq 2 + 4 + 4 + 4 = 14, \quad \|D^2 \mathcal{J}_S\| \leq 2 + 4 + 4 + 8 + 8 + 4 = 30. \quad (7.2)$$

Moreover

$$Dn_x = d_x^{-1/2} I - d_x^{-3/2} x \otimes x$$

and the exact second derivative has three terms with coefficient one and one term with coefficient three. Since $|x| \leq \sqrt{d_x}$,

$$\|Dn_x\| \leq d_x^{-1/2}, \quad \|D^2 n_x\| \leq (3 + 3)d_x^{-1} = 6d_x^{-1}. \quad (7.3)$$

Therefore

$$\boxed{\|D^2 F_S(x)\| \leq 14d_x^{-1/2}, \quad \|D^3 F_S(x)\| \leq 114d_x^{-1}.} \quad (7.4)$$

Combining (7.4) with (6.3) gives the floor-uniform conditional bounds

$$\boxed{\mathbb{E}\|D^2 F_S(X)\|^2 \leq \frac{49}{\lambda}, \quad \mathbb{E}\|D^3 F_S(X)\|^2 \leq \frac{3249}{2\lambda^2}.} \quad (7.5)$$

The next jet is genuinely critical already for the embedded production Pauli generator $S_3 = \text{diag}(1, 1, -1, -1, 0, 0)$. Take $x = (r, 0, y, 0, 0, 0)$. At zero floor the first component is

$$F_1(r, y) = r \frac{r^2 - y^2}{r^2 + y^2}, \quad \partial_y F_1(r, 0) = \frac{48}{r^3}. \quad (7.6)$$

At positive floor the same axis derivative is

$$\partial_y^4 F_{1,e}(r, 0) = \frac{24r(e + 2r^2)}{(e + r^2)^3}. \quad (7.7)$$

For $r \geq \sqrt{e}$, multiplication by r^3 gives $24(u+2)/(1+u)^3 \geq 9$ with $u = e/r^2 \in [0, 1]$. The rescaled derivative is continuous jointly in u and angle; compactness therefore supplies one fixed open cone about this Pauli axis and a constant $c > 0$, independent of e , on which $|D^4 F_{S_3,e}| \geq cr^{-3}$. Its squared six-dimensional Gaussian integral is bounded below by a fixed cone mass times

$$\int_{\sqrt{e}}^1 r^5 r^{-6} dr = \frac{1}{2} \log(1/e). \quad (7.8)$$

Uniform six-real ellipticity therefore repairs D^2 and D^3 , but not an L^2 fourth quotient jet. This rejects the elliptic-Gaussian fourth-jet repair, not fractional regularity or complete signed cancellation.

8. Conditional fractional exponent window and its missing spatial data

Pointwise ellipticity (6.1) controls inverse denominators but does not by itself control spatial fractional norms. Condition on all nonterminal fields and additionally assume the R-092 spatial increment, joint value–gradient moment, regression, Fubini, and Jensen hypotheses uniformly for the terminal Gaussian field and every production owner. Under these extra hypotheses the R-092 whole-product proof may be rerun with

$$p = 3, \quad q_H = 6, \quad \sigma = \frac{2}{3}, \quad \theta = \frac{3}{4}. \quad (8.1)$$

Indeed,

$$\frac{7}{12} < \sigma < \theta, \quad \frac{1}{2} = \frac{1}{p} + \frac{1}{q_H}, \quad p\theta = \frac{9}{4}, \quad 2p\theta = \frac{9}{2} < 6. \quad (8.2)$$

The normalized secant inequality

$$|\delta n| \leq 2 \min \left\{ 1, \frac{|a|}{\sqrt{d_-} + \sqrt{d_+}} \right\} \quad (8.3)$$

requires inverse moments only of orders $9/4$ and $9/2$, which Theorem 6.1 supplies. But the actual R-092 right side must also remain finite:

$$\mathbb{E}_\zeta \Re(\zeta)^2 < \infty, \quad \Re = N_+ \|Da\|_6 + \{D_\sigma + D_\infty(N_+ + N_-)\} \|Dx\|_6. \quad (8.4)$$

Only under this joint hypothesis does the parameter window support a conditional $W^{2/3,3} \times L^6 \rightarrow L^2$ estimate and hence outputs $\gamma < 2/3$, including $7/12$.

The extra hypothesis is indispensable. For independent standard Gaussians G_a, H_a , the six-component field $\zeta_N^a(x) = G_a \cos(Nx) + H_a \sin(Nx)$ has covariance I_6 at every point, while its $W^{\sigma,3}$ norm grows like N^σ . Thus pointwise ellipticity alone cannot imply a spatial bound.

R-104 includes a last independent base block additively in every endpoint of a fixed chart, so it may be disintegrated last if its pointwise covariance is uniformly positive. R-104 permits singular and shrinking source increments and does not prove the lower bound

$$\Delta C_K(x, x) \succeq \lambda_* I_6 \quad (8.5)$$

uniform in x , cutoff, chart, or directed refinement. It also does not prove that this common terminal disintegration intertwines with every root's trace, Wick forest, output projection, and conditioning. Existing data are upper trace estimates, not a six-component lower ellipticity theorem.

9. Conditional one-use square summation

Let $T_{A,j,k}$ be the R-088 sequential atoms after all same-root visits are recombined. Suppose that for integers $0 \leq k \leq j$ and $m \geq j + 5$,

$$\sum_A \int_0^\infty e^{-2t} \mathbb{E} \|\Pi_m P_t^{(j)} T_{A,j,k}\|_{\text{HS}}^2 dt \leq 2^{-2s(m-k)} q_k, \quad \sum_k q_k \leq Q. \quad (9.1)$$

Theorem 9.1 (conditional aggregate output ledger). If $0 < \gamma < s$, then

$$\boxed{\mathcal{B}_\gamma^{\text{out}} \leq \frac{2^{-10s}}{(1 - 2^{-s})^2 (1 - 2^{-2(s-\gamma)})} Q.} \quad (9.2)$$

Proof. For $0 < \eta < 2s$, weighted Cauchy–Schwarz gives

$$\left\| \sum_{k \leq j} x_k \right\|^2 \leq \frac{1}{1 - 2^{-\eta}} \sum_{k \leq j} 2^{\eta(j-k)} \|x_k\|^2. \quad (9.3)$$

Set $r = j - k \geq 0$ and $d = m - j \geq 5$, retain expectation inside, and use Tonelli. The two geometric series give

$$\frac{2^{-10s}}{(1 - 2^{-\eta})(1 - 2^{-2(s-\gamma)})(1 - 2^{\eta-2s})} Q. \quad (9.4)$$

The product of the first and third denominators is maximal at $\eta = s$: the two geometric ratios have fixed product 2^{-2s} and are equal there. This proves (9.2). Each q_k occurs once; no cross- k orthogonality is assumed. $\square$

At $s = 2/3$, $\gamma = 7/12$,

$$C_{2/3,7/12}^{\mathcal{B}} = \frac{2^{-20/3}}{(1 - 2^{-2/3})^2(1 - 2^{-1/6})} = 0.658881625872614\dots, \quad (9.5)$$

and the amplitude constant is

$$\sqrt{C_{2/3,7/12}^{\mathcal{B}}} = 0.811715236935106\dots \quad (9.6)$$

Hypotheses (9.1) are precisely the open production content for the R-088 sequential atoms. R-087 already proves an expectation-inside spatial estimate for its own self atom. What is missing is an owner-preserving identification or domination of each R-088 sequential atom by that R-087 atom, followed by the once-owned q_k sum.

10. Direct fixed-collar substitute and exact headroom

If only one fixed collar is needed, the literal $\mathcal{B}_{7/12}^{\text{out}}$ may be bypassed. The same summation at $s = \gamma = 7/12$ gives

$$\mathcal{H}_C(C) \leq D_{7/12} Q 2^{-7(C-5)/6}, \quad (10.1)$$

where

$$D_{7/12} = \frac{2^{-35/6}}{(1 - 2^{-7/12})^2(1 - 2^{-7/6})} = 0.285928885855479\dots, \quad (10.2)$$

and $\sqrt{D_{7/12}} = 0.534723186195885\dots$. This is still conditional on (9.1) and $\sum q_k \leq Q$.

Let the complete physical conversion factor be Λ_{own} and write

$$a_{\text{tail}}(C) \leq M 2^{-7(C-5)/12}, \quad M = \Lambda_{\text{own}} \sqrt{B_*}. \quad (10.3)$$

For a fixed three-channel block put

$$\sigma_0 = \frac{k^2}{d} < \min(e_0, f_0), \quad a_* = 2\sqrt{(e_0 - \sigma_0)(f_0 - \sigma_0)}. \quad (10.4)$$

Acceptance is exactly

$$a_0(C) + M 2^{-7(C-5)/12} < a_*. \quad (10.5)$$

If $a_0(C) \leq a_0 < a_*$ uniformly and $h_* = a_* - a_0$, the smallest sufficient integer is

$$C_{\min} = \max \left\{ 5, \left\lfloor 5 + \frac{12}{7} \log_2 \frac{M}{h_*} \right\rfloor + 1 \right\}. \quad (10.6)$$

The strict floor-plus-one is essential at equality.

Tail decay does not manufacture headroom. The logically admissible choice $a_*(C) - a_0(C) = M 2^{-7(C-5)/12}/2$ defeats (10.5) for every C . Production still needs either one chosen collar with certified positive headroom or a uniform $h_* > 0$ theorem. Relabelling (10.3) as the old entrywise C_{far} or C_{mix} is prohibited by R-133.

11. Why separate action payments diverge

Equation (4.3) is useful only if the forest and weighted current energy stay in one signed form. To see why, take the uncontrolled cutoff Gaussian with production covariance comparable to $\langle k \rangle^{-4} I$ in three spatial dimensions. Real-even covariance makes $W_\Lambda(x)$ and $\nabla W_\Lambda(x)$ independent at a point. Hence

$$\mathbb{E}|W_\Lambda(x)|^2 = O(1), \quad \mathbb{E}|\nabla W_\Lambda(x)|^2 \asymp \Lambda, \quad (11.1)$$

and

$$A_\Lambda^2 = \mathbb{E} \int |W_\Lambda|^2 |\nabla W_\Lambda|^2 \asymp \Lambda, \quad B_{e,\Lambda}^2 \asymp e\Lambda. \quad (11.2)$$

The source cost is zero and $\mathbb{E} \int |W_\Lambda|^6 = O(1)$. Therefore no cutoff-independent constants can make

$$A^2 + B_e^2 \lesssim 1 + X_{\text{src}} + Y_6 \quad (11.3)$$

hold by separate absolute estimates. The R-063 covariance-normal forest cancels exactly this derivative divergence. Taking an absolute value of each side before combining them discards that cancellation.

The next theorem must instead keep

$$\boxed{\mathfrak{F} - \left(\sqrt{\beta_{\text{op}}} A + \alpha \sqrt{c_1} B_e \right)^2 \geq -2\eta X_{\text{src}} - 2\zeta Y_6 - 2C,} \quad (11.4)$$

with $\eta < 9/20$, $\zeta < 3/20$, and every low, stationary, heat, trace, rational, Cartan, forest, source, and sextic owner retained once.

12. Absolute terminal packet and production scope

There is a narrow positive terminal identity. Suppose a final fresh Gaussian value g has no future feedback, is independent of the strict-past heat, has value-gradient independence, and its derivative covariance matches an independent trace probe G' . Pointwise let

$$W = W_0 + r + g, \quad V = V_0 + \partial g. \quad (12.1)$$

Conditioning on the value g makes $C_{6,e}(W)$ independent of ∂g . Cross terms vanish and matching covariance gives

$$\mathbb{E}_g \left[\|C_{6,e}(W)V\|^2 - \mathbb{E}_{G'} \|C_{6,e}(W)\partial G'\|^2 \right] = \mathbb{E}_g \|C_{6,e}(W)V_0\|^2 \geq 0. \quad (12.2)$$

Equation (12.2) does not replace the production moving-owner increment. The accepted fresh-shell calculation retains past covariance and stationary/low subtraction; already at $x = 0$, $Dx = 0$, and zero control it contains

$$-\frac{1}{2} \text{Tr}\{(P_{\Sigma_j} B(0))\Gamma_{j-1}\} < 0. \quad (12.3)$$

Earlier control roots also change the coefficient and derivative through all future feedback blocks,

$$D_{\xi_b} Z_h = S_b + \sum_{c>b} S_c D_b h_c, \quad (12.4)$$

so a raw trace probe does not automatically carry the Jacobian and connection. The R-104 same-root heat fixture $\mathbb{E}[G^2(G^2 - 1)] = 2$ independently detects the failure of automatic centering.

13. Evidence map and route decisions

Route	Verdict	Exact reason
Terminal elliptic smoothing	advanced conditionally	Sharp $q < 6$ negative moments control D^2/D^3 ; a local fractional 7/12 route additionally needs the R-092 joint spatial data, while R-104 has no uniform full-rank terminal covariance or owner intertwiner.
Pointwise-to-spatial transfer	failed	The covariance-one fields $G \cos(Nx) + H \sin(Nx)$ have fixed pointwise law but fractional norm of order N^σ .
Elliptic fourth jet	failed	The explicit $48r^{-3}$ component has logarithmically divergent six-dimensional L^2 norm.
Uniform finite-moment caps	failed	The bounded two-point family keeps every positive-moment cap below one while its Hessian is $-10c_1/(81\delta) + O(1)$; its exact mean does identify δ .
Floor removal at action level	advanced	Complete Pauli summation gives the sharp $\alpha\sqrt{c_1e}$ residual and one direct-sum owner inequality.
Separate $A^2 + B_e^2$ payment	failed	The uncontrolled covariance-normal Gaussian has linear derivative-energy divergence and bounded source/sextic.
Absolute terminal packet	advanced narrowly	Matching fresh derivative covariance is nonnegative, but past covariance and moving-owner terms remain.
Aggregate $\gamma = 7/12$	advanced conditionally	The geometric theorem has exact constant 0.6588816... under the still-open atom and one-use ledger.
Fixed-collar closure	inconclusive	Tail algebra is exact; near/balanced headroom, owner conversion, low constants, and the anchor remain absent.

The researcher-reusable decisions, including every failed route and its revisit condition, are stored as EXP-000514–EXP-000524. The fourth-jet, pointwise-to-spatial, and separate-payment failures are promoted to the negative registry. The earlier finite-moment-cap failure links to the already registered R-132 law-free result rather than duplicating it.

14. Devil’s-advocate review

Objection 1: the floor remainder carries a missing $\sqrt{3}$. DISMISSED. A rowwise triangle inequality would introduce it, but (3.4) sums the three Pauli squares exactly before estimating. The sharp coefficient is $\alpha^2 c_1 = 3/(320P)$, checked independently.

Objection 2: (4.3) proves the required physical Hessian. UPHELD. It is an action/value inequality. The two-point fixture has uniformly bounded values and moments but a divergent translation Hessian. No differentiation of (4.3) is used.

Objection 3: Gaussian smoothing removes every floor singularity. UPHELD. Only negative moments below the six-real codimension are uniform. Equation (7.6) realizes the critical fourth jet and (7.8) gives its logarithmic divergence.

Objection 4: R-104 already supplies the needed terminal Gaussian. VALID WITH MITIGATION. Every fixed chart has a last independent base block, but its covariance may be singular or shrink under refinement. The conditional theorems explicitly assume a uniform full-rank lower bound and owner-preserving disintegration.

Objection 5: (9.2) closes the one-use gate. UPHELD. The theorem assumes both the expectation-inside atom estimate and $\sum q_k \leq Q$. Those assumptions are the open production theorem, not a consequence of the geometric summation.

Objection 6: a large collar always creates acceptance. UPHELD. The near/balanced headroom may shrink faster than the tail. Equation (10.6) requires a separately certified positive h_* or one verified finite collar.

Objection 7: source and sextic can pay A^2 and B_e^2 separately. UPHELD. The zero-control Gaussian fixture (11.1)–(11.3) has bounded value/sextic and linear derivative energy. Only the signed covariance-normal forest-current combination can retain the necessary cancellation.

Objection 8: terminal positivity handles every root. UPHELD. Equation (12.2) is an absolute matching-covariance pointwise packet. The production moving-owner still contains (12.3), and earlier controls carry the future-feedback derivative (12.4).

15. Reproduction and remaining theorem

The primary executable derives every constant from the R-132/R-133 upstream artefacts and checks the Pauli identity, sharp floor residual, action Young form, two-point divergence, density constants, six-real moments, quotient jets, fourth-jet witness, fractional exponents, conditional shell constants, headroom fixture, and covariance derivative divergence. The independent standard-library executable re-derives the same constants without importing the primary module and uses separate Pauli samples and radial quadrature.

```
E:\Dev\TECT.venv\Scripts\python.exe
codes/foundations/a13_classii_terminal_smoothing_fixed_law_action_
aggregate-collar_boundary_verify.py
```

The next load-bearing theorem is not a separate forest lower bound followed by a separate current-energy estimate. It is the complete signed inequality (11.4), combined with an owner-preserving terminal-innovation reservation, joint value-gradient moments, shell-summable inverse-covariance costs, the production atom estimate (9.1), a one-use q_k ledger, and one certified near/balanced collar headroom. Only then may R-129 be applied to the low matrix and absolute anchor.

16. Result footer

Result ID: A13-CLASSII-TERMINAL-SMOOTHING-FIXED-LAW-ACTION-
AGGREGATE-COLLAR-BOUNDARY (R-134).

Claim: A13-CLASSII-RELATIVE-PHASE-SOURCE-BUDGET-OBSTRUCTION.

Precise statement: The complete six-row density-floor remainder has the sharp action bound $||C6e-C60|| \leq \alpha \sqrt{c1 e} |v|$, yielding a root/visit-free direct-sum forest-minus-variance inequality. Separate payment of its weighted current energies is cutoff-divergent. Under an added uniform six-real Gaussian covariance lower bound, negative moments $q < 6$ control D2 and D3 rational jets and conditionally repair a local $\gamma = 7/12$ fractional route; an explicit D4 jet diverges logarithmically.

A conditional root-atom theorem sums $B_{(7/12)}^{\text{out}}$ with square constant 0.658881625872614..., and a direct fixed-collar form has constant 0.285928885855479....

Scope: fixed finite cutoff and positive floor; complete recombined owners; action-level floor decomposition; assuming a uniformly elliptic six-real terminal innovation and the conditional root-atom and one-use q-ledger hypotheses stated above.

Dependencies: A1; R-063, R-087, R-088, R-091, R-092, R-104, R-116, R-119, R-123, R-125, R-128, R-129, R-130, R-131, R-132, and R-133.

Evidence grade: ANALYTIC, EXACT, EXECUTED.

Reproduction command: E:\Dev\TECT.venv\Scripts\python.exe
 codes/foundations/a13_classii_terminal_smoothing_fixed_law_action_aggregate_collar_boundary_verify.py

Expected output: R-134 integrated PASS; primary 44/44 and non-importing standard-library independent 35/35 children PASS; PDF and generated-surface gates PASS.

Falsification gate: failure of the Pauli sum, sharp floor constant, direct-sum variance inequality, two-point limit, density integrals, Gaussian negative-moment constants, D2/D3 jet factors, D4 witness, fractional exponents, conditional shell sums, direct collar constant, covariance derivative divergence, or any pinned authority, executable, JSON, exploration, negative, PDF, manifest, or public-surface assertion.

Tier before / after: T4 / T4; no promotion.

No-overclaim statement: no production-uniform terminal ellipticity or owner disintegration, signed forest-current estimate, one-use atom/q ledger, near/balanced headroom, low matrix, matching energy, strict gap, absolute anchor, either A13 gate, Nelson theorem, interacting measure, or Sector-A closure is proved.

Proof complete: false. Sector A closed: false.

Next required action: prove the combined signed forest-minus-weighted-current inequality with source and sextic used once; reserve a refinement-uniform full-rank terminal innovation compatible with every owner; prove the expectation-inside atom and one-use q ledger; certify one finite collar's near/balanced headroom; then apply R-129 once.

No-overclaim footer. R-134 proves a sharp action-level floor residual, an exact direct-sum owner inequality, a six-real Gaussian jet threshold, and conditional aggregate collar summations. It records the uniform-moment-cap, fourth-jet, separate-energy, and terminal-scope failures. It does not prove the production signed owner, one-use gate, headroom, anchor, either A13 gate, $\text{OVERLAP}_{\text{src}}$, Nelson, removals, an interacting measure, Sector A, or any T5–T7 promotion.